What Do Provably Powerful GNNs Learn? Interpreting Money-Laundering Detectors on an Expressiveness Ladder

Kristjan Egilsson
Stanford University
kde2804@stanford.edu

1 Motivation and research question

Money launderers move funds through recognizable transaction *patterns*: fan-ins, fan-outs, scatter-gathers, and cycles that return money to its origin. Message-passing GNNs are at most as expressive as the 1-WL test [Xu et al., 2019, Morris et al., 2019] and provably cannot count many such substructures [Chen et al., 2020]. Egressy et al. [2024] showed that three simple adaptations (reverse message passing, port numbering, and ego IDs) let a GNN detect any directed subgraph pattern in theory, and greatly improve laundering detection in practice. Performance gains alone do not show *how* the improved models work, yet a compliance analyst needs to know whether a flag reflects a laundering pattern or a shortcut.

Research question. *When an adaptation provably unlocks a pattern, does the trained model develop internal features that detect it, and does it rely on them?* Expressiveness theory says what a model *can* represent; neuron-level interpretability [Bau et al., 2017, Xuanyuan et al., 2023] and sparse dictionary learning [Bricken et al., 2023] reveal what it *does* represent. This project connects the two in an application with ground-truth pattern labels.

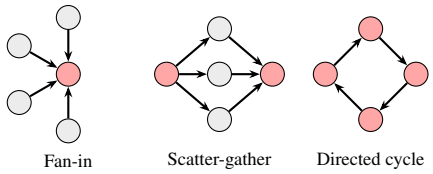
2 Application domain: datasets, task, and metrics

AMLworld, HI-Small [Altman et al., 2023]. A public synthetic transaction dataset from IBM's agent-based simulator, released on Kaggle under CDLA-Sharing-1.0. Accounts are nodes; transactions are directed, timestamped edges with amounts, currencies, and payment formats, forming a directed *multigraph*. HI-Small has about 515K accounts and 5M transactions over 10 days, roughly 1 in 981 of them laundering. Crucially, the release lists the laundering transactions belonging to each of 8 named patterns (fan-in, fan-out, gather-scatter, scatter-gather, cycle, bipartite, stack, random), which I will use as ground-truth *concepts*. **Task:** binary edge classification (is this transaction laundering?) with a chronological train/validation/test split, following Altman et al. [2023]. **Metrics:** minority-class F1 (the benchmark standard), area under the precision-recall curve, and per-pattern recall.

Controlled synthetic graphs. To validate the pipeline where every concept is known exactly, I will generate random directed multigraphs with planted patterns, following the synthetic tasks of Egressy et al. [2024]: node-level labels for degree thresholds, fan-in/fan-out, directed cycles of length 2–6, scatter-gather, and bicliques. I will release the generator and data with the Colab. **Optional real-data check:** the public Ethereum phishing network [Chen et al., 2021], where Egressy et al. [2024] also report gains.

Why this domain. In AML, the task-relevant structures are exactly the subgraph patterns expressiveness theory is about, ground-truth pattern labels exist, and explanations have direct regulatory value.

CS224W: Machine Learning with Graphs, Stanford University, Autumn 2026. Project proposal (Project 1: Application).



Model (cumulative)	Patterns it can detect
GIN	in-degree, edge statistics
+ reverse message passing	out-degree patterns
+ port numbering	fan-in, fan-out
+ ego IDs (Multi-GIN)	cycles, scatter-gather, bicliques

Figure 1: **Left:** three of the eight AMLworld laundering patterns (red: accounts to detect). **Right:** the expressiveness ladder and the patterns each added adaptation provably makes detectable [Egressy et al., 2024]; each row is a prediction to test.

3 Graph ML techniques

The expressiveness ladder. I will train four models that differ only in the adaptations of Egressy et al. [2024], built on GIN with edge features [Xu et al., 2019, Hu et al., 2020] using the public PyG [Fey and Lenssen, 2019] implementation (IBM `Multi-GNN`). With edge features e_{uv} and $\psi(h, e) = \mathrm{ReLU}(h + We)$, the layer for account v is

$$h_v^{(\ell+1)} = \mathrm{MLP}^{(\ell)}\Big((1+\epsilon)\, h_v^{(\ell)} + \sum_{u \in \mathcal{N}_{\mathrm{in}}(v)} \psi_{\mathrm{in}}\big(h_u^{(\ell)}, e_{uv}\big) + \sum_{w \in \mathcal{N}_{\mathrm{out}}(v)} \psi_{\mathrm{out}}\big(h_w^{(\ell)}, e_{vw}\big)\Big). \quad (1)$$

Plain GIN omits the last term; *reverse message passing* [Jaume et al., 2019] adds it, separating money received from money sent. *Port numbering* [Sato et al., 2019] appends to e_{uv} local IDs $(p_v^{\mathrm{in}}(u), p_u^{\mathrm{out}}(v))$ that each account assigns to its incoming and outgoing edges, so repeated transactions with the same counterparty can be told apart. *Ego IDs* [You et al., 2021] set $h_v^{(0)} = [x_v \,\|\, \mathbf{1}[v \in \mathcal{E}]]$, marking the endpoints $\mathcal{E}$ of the target transaction so the model can recognize paths that return to them. A transaction is scored as $\hat{y}_{uv} = \sigma\big(\mathrm{MLP}([h_u^{(L)} \,\|\, h_v^{(L)} \,\|\, e_{uv}])\big)$. Following the benchmark, I will use 2 layers and hidden size 64.

Interpretability methods. For each rung, I will extract hidden activations and run three analyses. (1) *Neuron dissection* [Bau et al., 2017, Xuanyuan et al., 2023]: for unit k with activation $a_k(v)$, let $M_k = \{v : a_k(v) > \tau_k\}$ with τ_k a top-quantile threshold, and for each concept mask C_c (pattern membership, degree bins, motif counts) compute $\mathrm{IoU}(k, c) = |M_k \cap C_c| / |M_k \cup C_c|$. Concept coverage for a model is $\max_k \mathrm{IoU}(k, c)$. (2) *Sparse autoencoders*: because GNN units are often polysemantic [Xuanyuan et al., 2023, Pertl et al., 2025], I will train a TopK SAE [Gao et al., 2024], $z = \mathrm{TopK}\big(W_{\mathrm{enc}}(h - b)\big)$, $\hat{h} = W_{\mathrm{dec}} z + b$, minimizing $\|h - \hat{h}\|_2^2$, and repeat the dissection on SAE latents. (3) *Causal ablation*: replace a unit or latent with its mean and measure the drop in recall on each pattern, $\Delta_c(k) = \mathrm{Rec}_c - \mathrm{Rec}_c^{(-k)}$. A feature counts as a detector only if ablating it hurts that pattern specifically. Linear probes [Akhondzadeh et al., 2023] give a decodability baseline.

Hypotheses. H1: concept coverage grows along the ladder as Figure 1 predicts (e.g., cycle detectors appear only with ego IDs). **H2:** new detectors are causally used, with pattern-specific ablation effects. **H3:** where no single neuron detects a provably detectable pattern, SAE latents recover it. Disagreements with theory are findings in their own right.

Why these models fit the data. Transaction graphs are directed multigraphs whose laundering signal lives in patterns plain message passing provably misses; Multi-GIN is built for this setting and is a strong AMLworld baseline [Egressy et al., 2024]. One adaptation per rung isolates one capability at a time, making the analysis a controlled experiment.

4 Relation to prior work and plan

Prior work has found concept-detecting neurons in standard GNNs [Xuanyuan et al., 2023], measured superposition across GCN, GIN, and GAT [Pertl et al., 2025], and probed what architectures encode [Akhondzadeh et al., 2023, Pelletreau-Duris et al., 2025]. To my knowledge, none studies beyond-1-WL architectures with rung-specific predictions, tests causal use, applies sparse autoencoders to message-passing GNNs, or uses financial transaction graphs.

Plan (solo). *Milestone (Nov. 10):* data pipelines, all four rungs trained on both datasets, dissection working on synthetic data. *Nov. 11–28:* dissection, probes, SAEs, and ablations on AMLworld. *Dec. 1–10:* Ethereum check, blog, Colab. **Risks:** with limited compute I will fix hyperparameters; if neurons are uninterpretable, SAEs lead.

References

M. S. Akhondzadeh, V. Lingam, and A. Bojchevski. Probing graph representations. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 206 of *PMLR*, 2023.

E. Altman, J. Blanuša, L. von Niederhäusern, B. Egressy, A. Anghel, and K. Atasu. Realistic synthetic financial transactions for anti-money laundering models. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023.

D. Bau, B. Zhou, A. Khosla, A. Oliva, and A. Torralba. Network dissection: Quantifying interpretability of deep visual representations. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023.

L. Chen, J. Peng, Y. Liu, J. Li, F. Xie, and Z. Zheng. Phishing scams detection in Ethereum transaction network. *ACM Transactions on Internet Technology*, 21(1):1–16, 2021.

Z. Chen, L. Chen, S. Villar, and J. Bruna. Can graph neural networks count substructures? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

B. Egressy, L. von Niederhäusern, J. Blanuša, E. Altman, R. Wattenhofer, and K. Atasu. Provably powerful graph neural networks for directed multigraphs. In *AAAI Conference on Artificial Intelligence*, volume 38, pages 11838–11846, 2024.

M. Fey and J. E. Lenssen. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

L. Gao, T. Dupré la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, and J. Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.

W. Hu, B. Liu, J. Gomes, M. Zitnik, P. Liang, V. Pande, and J. Leskovec. Strategies for pre-training graph neural networks. In *International Conference on Learning Representations (ICLR)*, 2020.

G. Jaume, A.-p. Nguyen, M. R. Martínez, J.-P. Thiran, and M. Gabrani. edGNN: A simple and powerful GNN for directed labeled graphs. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*, 2019.

C. Morris, M. Ritzert, M. Fey, W. L. Hamilton, J. E. Lenssen, G. Rattan, and M. Grohe. Weisfeiler and Leman go neural: Higher-order graph neural networks. In *AAAI Conference on Artificial Intelligence*, 2019.

T. Pelletreau-Duris, R. van Bakel, and M. Cochez. Do graph neural network states contain graph properties? In *Conference on Neurosymbolic Learning and Reasoning (NeSy)*, volume 284 of *PMLR*, 2025.

L. Pertl, H. Xuanyuan, and P. Liò. Superposition in graph neural networks. *arXiv preprint arXiv:2509.00928*, 2025.

R. Sato, M. Yamada, and H. Kashima. Approximation ratios of graph neural networks for combinatorial problems. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

K. Xu, W. Hu, J. Leskovec, and S. Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations (ICLR)*, 2019.

H. Xuanyuan, P. Barbiero, D. Georgiev, L. C. Magister, and P. Liò. Global concept-based interpretability for graph neural networks via neuron analysis. In *AAAI Conference on Artificial Intelligence*, pages 10675–10683, 2023.

J. You, J. M. Gomes-Selman, R. Ying, and J. Leskovec. Identity-aware graph neural networks. In *AAAI Conference on Artificial Intelligence*, 2021.